

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year 2019 – 2020 (Even Semester)

Innovative practice(s) description

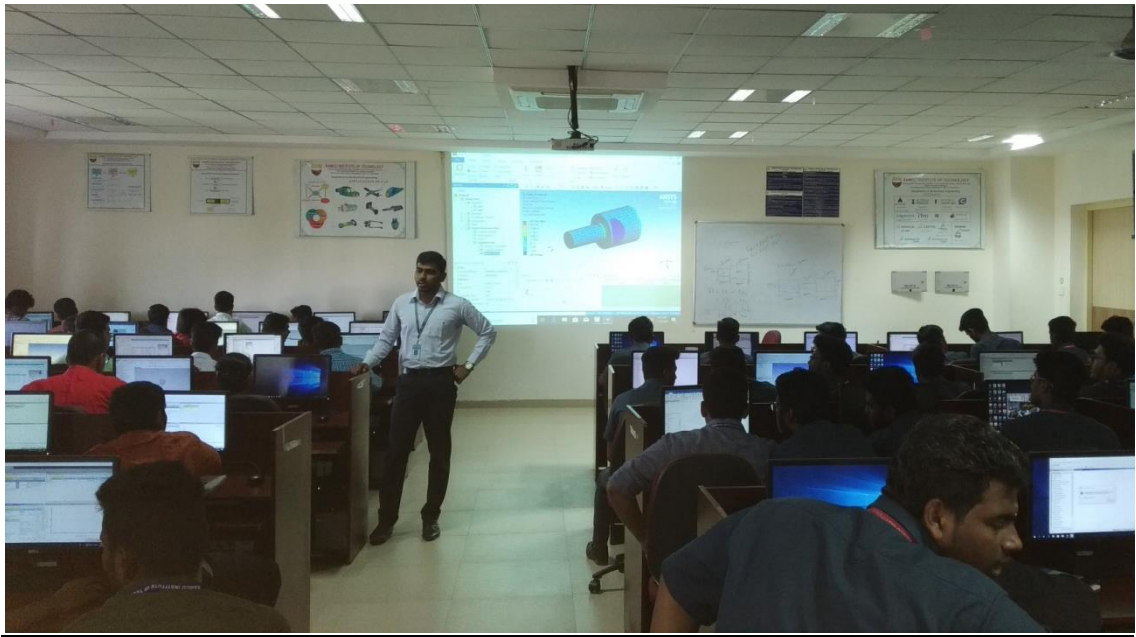
Course code and title : CE8395 Strength of Materials for Mechanical Engineers

Name of the Instructor(s): Mr.T.Selva Sundar, AP/Mechanical

Mr.K.Amudhan, AP/Mechanical

Date : 16.02.2020

Use of Simulation tools to verify analytical results and for Visualizing the physical process



In the Unit-I (**Stress, Strain and deformation of solids**) students come across the concepts of deformation of simple and compound members subjected to tensile and compressive loading. The analytical concepts are explained in classroom and relevant problems are solved. To visualize these concepts students were taken to mechanical CAD laboratory and using ANSYS software package the problem which was solved in the classroom was modelled and simulated. The analytical results and the numerical results were compared; in addition to that deformation and stress plots are also generated.

OUTCOME:

Students were also able to correlate with the problems, formulas that they learn, apply while solving problems in the classroom and the simulation what they perform in the Laboratory. The students will be able to compare the concepts of deformation, stress and strain in simple and compound bars. Students also gained hands on experience on working with ANSYS static structural module.